Why is OZONE LAYER so Important ?

STRATOSPHERIC OZONE - IMPORTANCE

Fauna & Human

Control UVB-entry into earth atmosphere; laboratory & epidemiologist found that:-

- UVB causes **Non-Melanoma Skin Cancer.**
- Promotes Malignant **Melanoma Development.**
- Is linked with the **Development of Cataracts** (a clouding of eye's lens).

Flora

Plants have abilities to adapt to high UVB-levels, still they're directly affected ---

- **Indirect UVB induced changes-** in **plant form, nutrient-distribution, developmental-phases timing, secondary metabolism.**
- Effects **plant's competitive balance, herbivory, plant-diseases & biogeochemical changes.**

Marine Ecosystems

Phytoplankton's (foundation of aquatic food webs) productivity being limited to euphotic zone, sunlight having UVB Radiation affects their **orientation & motility** , causing threat to aquatic lives.

UVB-Radiation causes damage to **Early-Developmental stages** of many marine animals, e.g. reduced reproductive capacity, impaired larval development.

Bio-geochemical Cycles

UVB affects **terrestrial & aquatic Biogeochemical Cycles**, altering both sources & sinks of Greenhouse & chemically important Trace-gases **(e.g. CO2, CO, COS, O 3, etc.)**

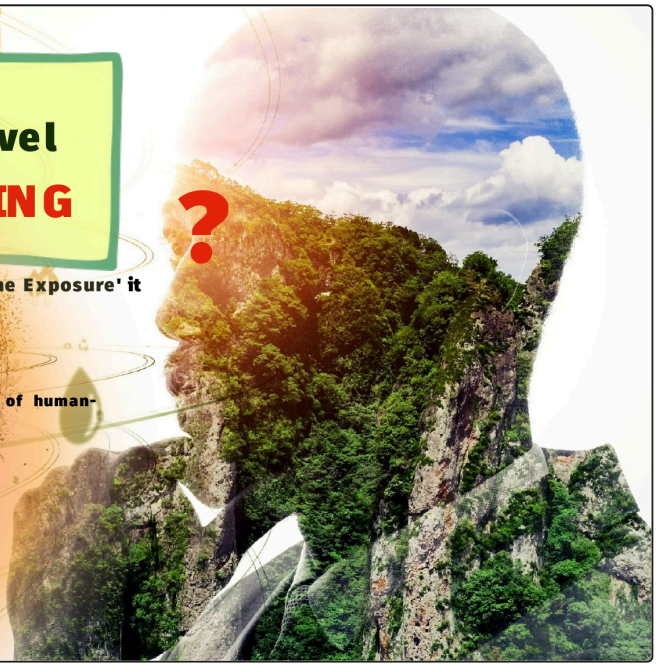
These potential changes contributing to biosphere-atmosphere feedbacks, mitigates or amplifies the atmospheric concentrations of these gases.

Materials

Synthetic Polymers, natural bio-polymers, commercially sold materials are highly damaged and their quality & longevity is affected by UVB Radiation.

WHY IS Ground level OZONE LIFE-THREATENING ?

TROPOSPHERIC OZONE EXPOSURE



Depending on the level of 'Troposhepric Ozone Exposure' it is detrimental in several ways :

- **Hinders vigorous and deep breathing; pain while breathing deeply**
- **Inflammation & irritation of the tissue-linings of human-airways**
- **Lungs turn more susceptible to infection**
- **Aggravate lung diseases like-asthma, emphysems & chronic bronchitis**
- **Increases frequency of Asthma-Attacks**
- **Ozone causes airway-muscles to constrict, trapping air in the alveoli- leading to wheezing & shortness of breath**

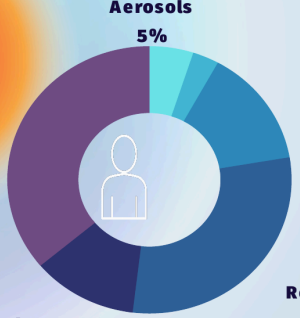
HUMAN-ACTIONS BEHIND OZONE-LAYER CHANGES

ANTHROPOGENIC OZONE-LAYER CHANGES

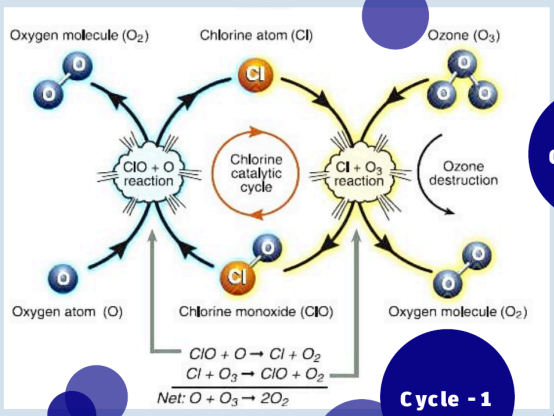
Human Civilization Growth has always prioritized comfort over Environmental-Balance : this self-centered attitude can actually threaten the future-generations and survival of other species.

Different Industrial Emissions, household use of refrigerators producing **CFCs**, automobile engines generating harmful **Greenhouse-Gases (CFCs & HCFCs)** that form **Chlorine & Bromine free-radicals** that are mainly responsible for breaking down thus destroying the ozone-layer.

O_3 to O_2



Cycles of Ozone-Layer Depletion



Cycle - 1

Cycle - 2

$$ClO + ClO \rightarrow (ClO)_2$$
$$(ClO)_2 + \text{sunlight} \rightarrow ClOO + Cl$$
$$ClOO \rightarrow Cl + O_2$$
$$2(Cl + O_3 \rightarrow ClO + O_2)$$
$$\text{Net: } 2O_3 \rightarrow 3O_2$$

or

$$ClO + BrO \rightarrow Cl + Br + O_2$$

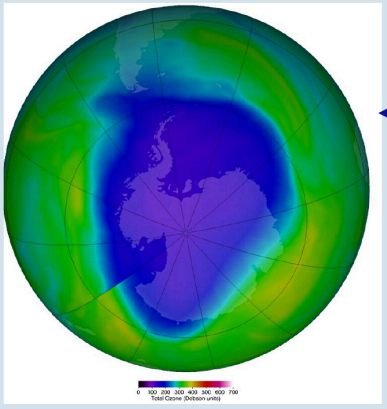
$$\begin{pmatrix} ClO + BrO \rightarrow BrCl + O_2 \\ BrCl + \text{sunlight} \rightarrow Cl + Br \end{pmatrix}$$

Cycle - 3

$$Cl + O_3 \rightarrow ClO + O_2$$
$$Br + O_3 \rightarrow BrO + O_2$$
$$\text{Net: } 2O_3 \rightarrow 3O_2$$

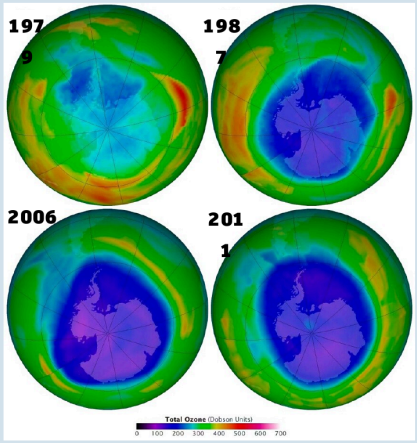
THE OZONE-HOLE

A MOST RECENT DATA

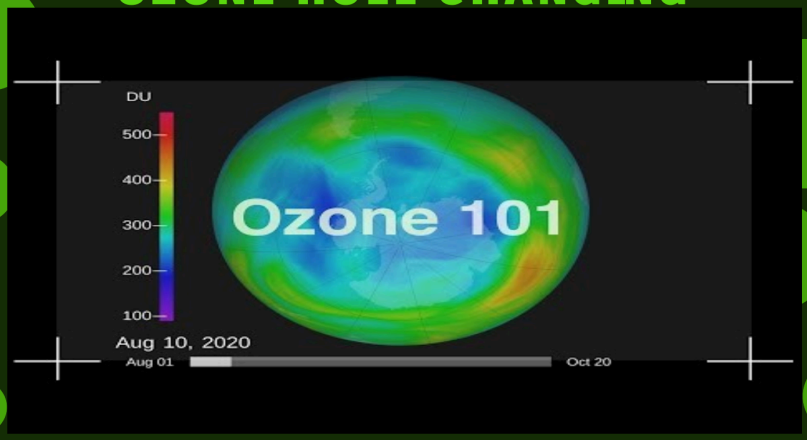


3 November, 2022

False-color view of total ozone over the Antarctic pole. The purple and blue colours are where there is the least ozone, and the yellows and reds are where there is more ozone

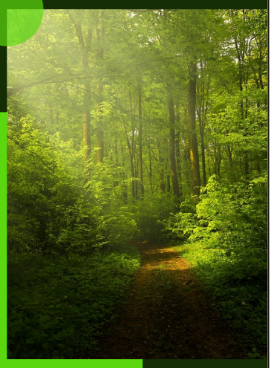


OZONE-HOLE CHANGING



CORONA PANDEMIC OBSERVATIONS

CERTAIN INDUSTRIAL-EMISSIONS CONTRIBUTE TO O_3 - FORMATION





Written on June 9, 2021, NASA found -
"Local lockdowns brought fast global ozone-reductions

Emissions of nitrogen oxides (NOx) – which create ozone, a danger to human health and to climate – decreased 15% globally, with local reductions as high as 50%

As a result of the lower NOx emissions, by June 2020, global ozone levels had dropped to a level that policy makers thought would take at least 15 years to reach by conventional means, such as regulations.



INTERNATIONAL PROTOCOLS

INTERNATIONAL PROTOCOLS ADOPTED FOR OZONE PROTECTION

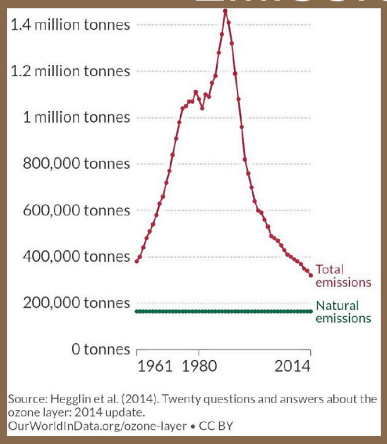
- The Vienna Convention - 1985
- The Montreal Protocol - 1987

Amendments to the Montreal Protocols:

- **The London Amendment - 1990**
- **The Copenhagen Amendment - 1992**
- **The Montreal Amendment - 1997**
- **The Beijing Amendment - 1999**
- **The Kigali Amendment - 2016**



EMISSIONS : RISE & FALL



Source: Hegglin et al. (2014). Twenty questions and answers about the ozone layer: 2014 update.
OurWorldInData.org/ozone-layer • CC BY

In the chart beside, emissions of Ozone-depleting substances are seen from 1960 onwards, measured in tons of CFC-11 equivalents.

CFC-11 is a standardized measure to normalize the sum of a range of ozone-depleting substances, to a value equivalent to their potential to deplete ozone.

Here, a clear growth-peak-reduction is seen with also a rapid rise by 3 folds from 1960s to 1980s.

By 2010, emissions returned to 1960 levels a result of various strict international regulatory agreements